in_valid

rst_n

clk

in_couple[1:0]

clk_IBUF_inst

IBUF I O

FSM_sequential_state_reg[0]_i_1

LUT3 O

I0
I1

I2clk_IBUF_BUFG_inst BUFG I O

FSM_sequential_state_reg_reg[0]

FDRE Q C

CEDR

in_valid_IBUF_inst

IBUF I O

count[10]_i_3

LUT6 O

I0
I1
I2
I3
I4
I5

count[5]_i_2

LUT5 O

I0
I1
I2
I3
I4

count[9]_i_2

LUT6 O

I0
I1
I2
I3
I4
I5

count[10]_i_4

LUT4 O

I0
I1
I2
I3

count[0]_i_1

LUT6 O

I0
I1
I2
I3
I4

I5count[1]_i_1

LUT6 O

I0
I1
I2
I3
I4

I5count[2]_i_1

LUT4 O

I0
I1
I2

I3count[3]_i_1

LUT5 O

I0
I1
I2
I3

I4count[4]_i_1

LUT6 O

I0
I1
I2
I3
I4

I5count[5]_i_1

LUT4 O

I0
I1
I2

I3count[6]_i_1

LUT5 O

I0
I1
I2
I3

I4count[7]_i_1

LUT6 O

I0
I1
I2
I3
I4

I5 ce_load_init_i_1

LUT5 O

I0
I1
I2
I3
I4

count[10]_i_2

LUT4 O

I0
I1
I2
I3

count[8]_i_1

LUT5 O

I0
I1
I2
I3
I4

count[9]_i_1

LUT6 O

I0
I1
I2
I3
I4
I5

count_reg[0]

FDRE Q C

CEDR

count_reg[1]

FDRE Q C

CEDR

count_reg[2]

FDRE Q C

CEDR

count_reg[3]

FDRE Q C

CEDR

count_reg[4]

FDRE Q C

CEDR

count_reg[5]

FDRE Q C

CEDR

count_reg[6]

FDRE Q C

CEDR

count_reg[7]

FDRE Q C

CEDR

count_reg[8]

FDRE Q C

CEDR

count_reg[9]

FDRE Q C

CEDR

count_reg[10]

FDRE Q C

CEDR

ce_init_state_reg[3]

FDRE Q C

CEDR

ce_load_init_reg

FDRE Q C

CEDR

in_couple_IBUF[0]_inst

IBUF I O

arp_swap_q_reg

FDRE Q C

CEDR

in_couple_IBUF[1]_inst

IBUF I O

ce_init_state_reg[0]

FDRE Q C

CEDR

ce_init_state_reg[1]

FDRE Q C

CEDR

ce_init_state_reg[2]

FDRE Q C

CEDR

pipe_valid_reg

FDRE Q C

CEDR

rst_n_IBUF_inst

IBUF I O

u_sys_ram

ram_sdp_1

FSM_sequential_state_reg_reg[2]

clk_IBUF_BUFG

in_valid_IBUF mem_reg_i_6_0

state_reg_reg[3]_0
state_reg_reg[3]_2

ADDRBWRADDR[5:0]
D[1:0]

DIADI[1:0] Q[10:0] mem_reg_0[3:0] ce_init_state_reg[3] state_reg_reg[3] state_reg_reg[3]_1[1:0]

u_ce

constituent_encoder

FSM_sequential_state_reg_reg[2]

clk_IBUF_BUFG

in_valid_IBUF mem_reg
rst_n_IBUF
state_reg_reg[0]_0
state_reg_reg[0]_1

D[3:0] DIADI[1:0] Q[1:0] SR

ce_init_state_reg[0][3:0]

state_reg_reg[2]_0[2:0] state_reg_reg[3]_0

arp_start_i_1

LUT3 O

I0
I1

I2 arp_enable_reg

FDRE Q C

CEDR

arp_start_reg

FDRE Q C

CEDR

FSM_sequential_state_reg[1]_i_1

LUT3 O

I0
I1

I2 FSM_sequential_state_reg[2]_i_1
LUT3 O

I0
I1
I2

FSM_sequential_state_reg[3]_i_2

LUT4 O

I0
I1
I2
I3

u_arp

arp_interleaver

FSM_sequential_state_reg_reg[0]_0
active_reg_reg_1

active_reg_reg_2
active_reg_reg_3
arp_swap_q_reg

clk_IBUF_BUFG count_reg[2] count_reg[3] count_reg[6]

in_valid_IBUF
j_reg_reg[0]_0

pipe_valid_reg
rst_n_IBUF

ADDRBWRADDR[5:0] E
FSM_sequential_state_reg_reg[0] Q[3:0] SR
active_reg_reg_0
ce_load_init_i_2_0[10:0]

par_count[8]_i_2

LUT6 O

I0
I1
I2
I3
I4
I5

FSM_sequential_state_reg_reg[1]

FDRE Q C

CEDR FSM_sequential_state_reg_reg[2]
FDSE Q C

CEDS

FSM_sequential_state_reg_reg[3]

FDRE Q C

CEDR

par_count[10]_i_3

LUT3 O

I0
I1

I2 par_count[4]_i_2

LUT4 O

I0
I1
I2

I3 par_count[10]_i_4

LUT4 O

I0
I1
I2

I3 par_count[5]_i_2

LUT5 O

I0
I1
I2
I3
I4

par_count[0]_i_1

LUT5 O

I0
I1
I2
I3

I4 par_count[1]_i_1

LUT6 O

I0
I1
I2
I3
I4

I5 par_count[2]_i_1

LUT5 O

I0
I1
I2
I3

I4 par_count[3]_i_1

LUT6 O

I0
I1
I2
I3
I4

I5 par_count[4]_i_1

LUT6 O

I0
I1
I2
I3
I4

I5 par_count[5]_i_1

LUT6 O

I0
I1
I2
I3
I4

I5 par_count[10]_i_2

LUT5 O

I0
I1
I2
I3

I4 par_count[6]_i_1

LUT6 O

I0
I1
I2
I3
I4

I5 par_count[7]_i_1

LUT5 O

I0
I1
I2
I3

I4 par_count[8]_i_1

LUT6 O

I0
I1
I2
I3
I4

I5 par_count[9]_i_1

LUT6 O

I0
I1
I2
I3
I4
I5

par_count_reg[10]_psbram_inv LUT1

I0 O

par_count_reg[10]_psbram_inv_1
LUT1

I0 O

par_count_reg[0]

FDRE Q C

CEDR

par_count_reg[1]

FDRE Q C

CEDR

par_count_reg[2]

FDRE Q C

CEDR

par_count_reg[3]

FDRE Q C

CEDR

par_count_reg[4]

FDRE Q C

CEDR

par_count_reg[5]

FDRE Q C

CEDR

par_count_reg[6]

FDRE Q C

CEDR

par_count_reg[7]

FDRE Q C

CEDR

par_count_reg[8]

FDRE Q C

CEDR

par_count_reg[9]

FDRE Q C

CEDR

par_count_reg[10]

FDRE Q C

CEDR

par_count_reg[10]_psbram

FDRE Q C

CEDR

par_count_reg[10]_psbram_1

FDRE Q C

CEDR

par_count_reg[10]_psbram_2

FDRE Q C

CEDR

par_count_reg[10]_psbram_3

FDRE Q C

CEDR

out_done_i_1

LUT4 O

I0
I1
I2
I3

out_valid_i_1

LUT5 O

I0
I1
I2
I3

I4 out_valid_i_2
LUT2

I0 O
I1

in_ready_r_i_1

LUT4 O

I0
I1
I2
I3

par1_ram

ram_sdp

DOBDO[0]_alias DOBDO[1]_alias

clk_IBUF_BUFG

lopt mem_reg_2

...count_reg[10]_psbram_2_psbram_n_alias par_count_reg[10]_psbram_n_1_alias

DIADI[1:0]E DOBDO[1:0] Q[10:0] mem_reg_0[10:0] mem_reg_1[2:0]

par2_ram

ram_sdp_0

DOBDO[0]_alias DOBDO[1]_alias

clk_IBUF_BUFG

lopt mem_reg_2

par_count_reg[10]_psbram_n_alias par_count_reg[10]_psbram_psbram_n_alias

DIADI[1:0] DOBDO[1:0] Q[10:0] E mem_reg_0[10:0] mem_reg_1[3:0]
out_done_reg

FDRE Q C

CEDR

out_valid_reg

FDRE Q C

CEDR

in_ready_r_reg

FDPE Q C

CED

PRE out_symbol_reg[4]

FDRE Q C

CEDR

out_symbol_reg[5]

FDRE Q C

CEDR

out_done_OBUF_inst OBUF I O

out_valid_OBUF_inst OBUF I O
out_symbol_OBUF[0]_inst OBUF I O
out_symbol_OBUF[1]_inst OBUF I O
in_ready_OBUF_inst OBUF I O
out_symbol_OBUF[2]_inst OBUF I O
out_symbol_OBUF[3]_inst OBUF I O
out_symbol_OBUF[4]_inst OBUF I O

out_symbol_OBUF[5]_inst OBUF I O

out_done

out_valid

in_ready out_symbol[5:0]

arp_Q0[10:0] arp_Q1[10:0] arp_Q2[10:0] arp_Q3[10:0] arp_p[10:0] blk_size_N[10:0]